

Cluster Validation Statistics (updated June 2025)

DANA 4840

```
library(factoextra)

## Warning: package 'factoextra' was built under R version 4.3.3

## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.3.3

## Welcome! Want to learn more? See two factoextra-related books at
https://goo.gl/ve3WBa

library(fpc)

## Warning: package 'fpc' was built under R version 4.3.3
```

We'll use the built-in R data set iris:

```
# Excluding the column "Species" at position 5
df <- iris[, -5]
# Standardize
df <- scale(df)
```

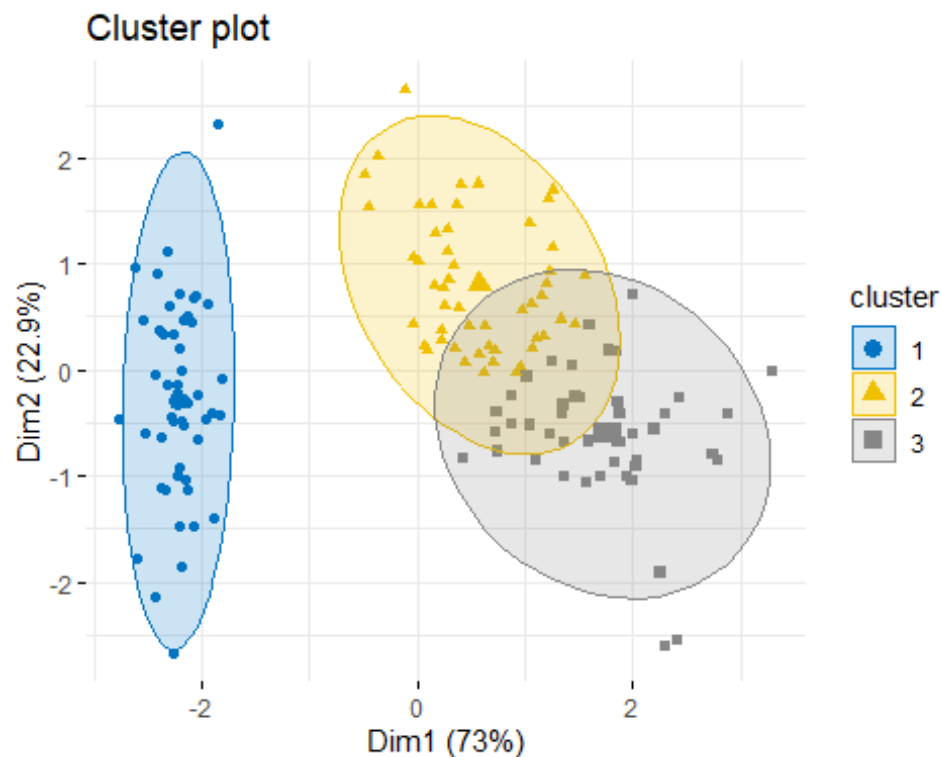
Clustering Analysis

We'll use the function eclust() [enhanced clustering, in factoextra] which provides several advantages:

- It simplifies the workflow of clustering analysis
- It can be used to compute hierarchical clustering and partitioning clustering in a single line function call
- Compared to the standard partitioning functions (kmeans, pam, clara and fanny) which requires the user to specify the optimal number of clusters, the function eclust() computes automatically the gap statistic for estimating the right number of clusters.
- It provides silhouette information for all partitioning methods and hierarchical clustering
- It draws beautiful graphs using ggplot2

To compute a partitioning clustering, such as k-means clustering with $k = 3$, type this:

```
# K-means clustering
km.res <- eclust(df, "kmeans", k = 3, nstart = 25, graph = FALSE)
# Visualize k-means clusters
fviz_cluster(km.res, geom = "point", ellipse.type = "norm",
              palette = "jco", ggtheme = theme_minimal())
```

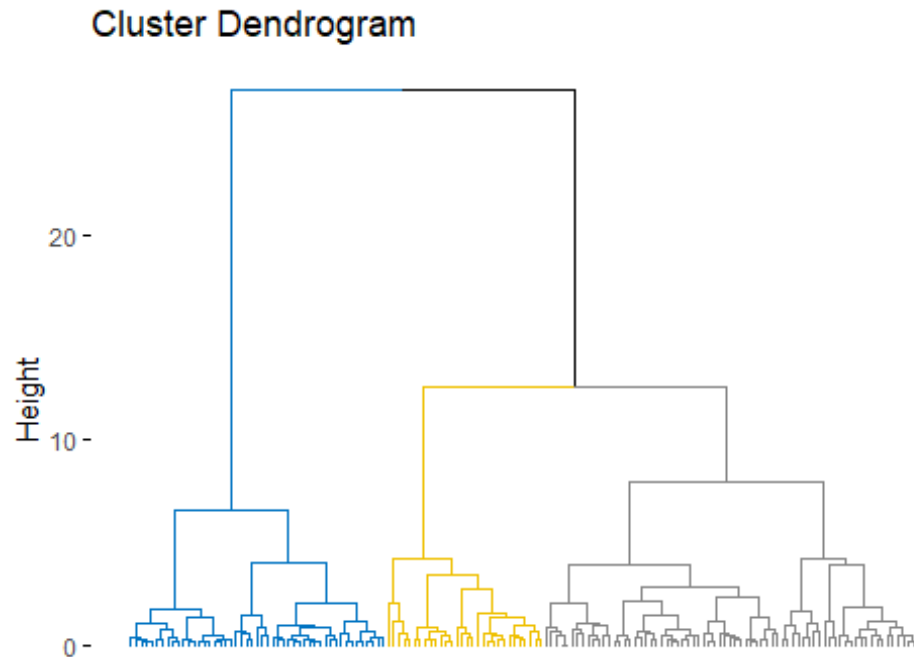


To compute a hierarchical clustering, use this:

```
# Hierarchical clustering
hc.res <- eclust(df, "hclust", k = 3, hc_metric = "euclidean",
                hc_method = "ward.D2", graph = FALSE)

# Visualize dendrograms
fviz_dend(hc.res, show_labels = FALSE,
          palette = "jco", as.ggplot = TRUE)

## Warning: The `<scale>` argument of `guides()` cannot be `FALSE`. Use
"none" instead as
## of ggplot2 3.3.4.
## [i] The deprecated feature was likely used in the factoextra package.
## Please report the issue at
<https://github.com/kassambara/factoextra/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



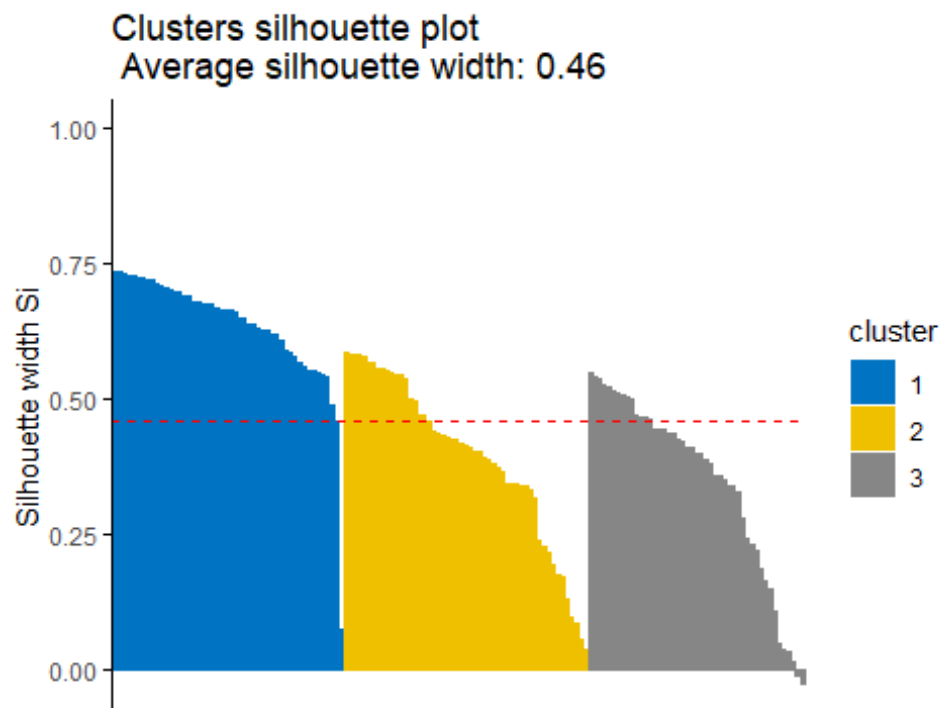
Cluster Validation

Recall that the silhouette coefficient (S_i) measures how similar an object i is to the other objects in its own cluster versus those in the neighbor cluster. S_i values range from 1 to -1:

- A value of S_i close to 1 indicates that the object is well clustered. In other words, the object i is similar to the other objects in its group.
- A value of S_i close to -1 indicates that the object is poorly clustered, and that assignment to some other cluster would probably improve the overall results.

```
fviz_silhouette(km.res, palette = "jco",  
                ggtheme = theme_classic())
```

```
## cluster size ave.sil.width  
## 1      1    50          0.64  
## 2      2    53          0.39  
## 3      3    47          0.35
```



Silhouette information can be extracted as follow:

```
## # Silhouette information
silinfo <- km.res$silinfo
names(silinfo)

## [1] "widths"          "clus.avg.widths" "avg.width"

## # Silhouette widths of each observation
head(silinfo$widths[, 1:3], 10)

##   cluster neighbor sil_width
## 1        1         2 0.7341949
## 41       1         2 0.7333345
## 8        1         2 0.7308169
## 18       1         2 0.7287522
## 5        1         2 0.7284741
## 40       1         2 0.7247047
## 38       1         2 0.7244191
## 12       1         2 0.7217939
## 28       1         2 0.7215103
## 29       1         2 0.7145192

## # Average silhouette width of each cluster
silinfo$clus.avg.widths

## [1] 0.6363162 0.3933772 0.3473922
```

```
## # The total average (mean of all individual silhouette widths)
silinfo$avg.width

## [1] 0.4599482

## # The size of each clusters
km.res$size

## [1] 50 53 47
```

It can be seen that several samples, in cluster 3, have a negative silhouette coefficient. This means that they are not in the right cluster. We can find the name of these samples and determine the clusters they are closer (neighbor cluster), as follow:

```
# Silhouette width of observation
sil <- km.res$silinfo$widths[, 1:3]
# Objects with negative silhouette
neg_sil_index <- which(sil[, 'sil_width'] < 0)
sil[neg_sil_index, , drop = FALSE]

##      cluster neighbor  sil_width
## 112         3         2 -0.01058434
## 128         3         2 -0.02489394
```

Computing Dunn index and other cluster validation statistics

In the following sections, we'll compute the clustering quality statistics for k-means. Look at the within.cluster.ss (within clusters sum of squares), the average.within (average distance within clusters) and clus.avg.silwidths (vector of cluster average silhouette widths).

```
# Statistics for k-means clustering
km_stats <- cluster.stats(dist(df), km.res$cluster)
# Dun index
km_stats$dunn

## [1] 0.02649665
```

External Clustering Validation

Among the values returned by the function cluster.stats(), there are indexes to assess the similarity of two clustering. One of the indices is the corrected Rand index.

We know that the iris data contains exactly 3 groups of species.

Does the K-means clustering matches with the true structure of the data?

We can use the function cluster.stats() to answer to this question.

Let start by computing a cross-tabulation between k-means clusters and the reference Species labels:

```
table(iris$Species, km.res$cluster)
```

```
##
##           1  2  3
##  setosa    50  0  0
##  versicolor 0 39 11
##  virginica  0 14 36
```

It can be seen that: - All setosa species (n = 50) has been classified in cluster 1 - A large number of versicolor species (n = 39) has been classified in cluster 3. Some of them (n = 11) have been classified in cluster 2. - A large number of virginica species (n = 36) has been classified in cluster 2. Some of them (n = 14) have been classified in cluster 3.

It's possible to quantify the agreement between Species and k-means clusters using the corrected Rand index:

```
#Library("fpc")
# Compute cluster stats
species <- as.numeric(iris$Species)
clust_stats <- cluster.stats(d = dist(df),
                             species, km.res$cluster)

# Corrected Rand index
clust_stats$corrected.rand

## [1] 0.6201352
```

The corrected **Rand index** provides a measure for assessing the similarity between two partitions, adjusted for chance. Its range is -1 (no agreement) to 1 (perfect agreement). Agreement between the species types and the cluster solution is 0.62 using **Rand index**.

The same analysis can be computed for both PAM and hierarchical clustering:

```
## # Agreement between species and pam clusters
pam.res <- eclust(df, "pam", k = 3, graph = FALSE)
table(iris$Species, pam.res$cluster)

##
##           1  2  3
##  setosa    50  0  0
##  versicolor 0  9 41
##  virginica  0 36 14

cluster.stats(d = dist(df),
              species, pam.res$cluster)$corrected.rand

## [1] 0.6416027

## # Agreement between species and HC clusters
res.hc <- eclust(df, "hclust", k = 3, graph = FALSE)
table(iris$Species, res.hc$cluster)

##
##           1  2  3
```

```
## setosa      49  1  0
## versicolor  0 27 23
## virginica   0  2 48

cluster.stats(d = dist(df),
              species, res.hc$cluster)$corrected.rand

## [1] 0.615323
```

External clustering validation, can be used to select suitable clustering algorithm for a given data set.